

Mid-term problem statement

Target application: Remote control for electric skateboard

Components required:

1. Arduino Uno
2. Breadboard power supply
3. Joystick
4. 7 seg display
5. Shift register
6. Accelerometer
7. Ultrasound distance sensor
8. Servo motor
9. DC motor
10. Stepper motor through motor driver
11. Photoresistor
12. Potentiometer

Functionalities:

1. When you tilt forward the remote (accelerometer) the speed of the skateboard increases (DC motor). When you keep the remote horizontal the speed stays the same. When you tilt it back, the skateboard slows down (brakes). [Mandatory, 10 points]
2. The speed of the skateboard is continuously displayed (on the 7 segment display, connected by through a shift register). We will use arbitrary units for the speed; 9 for max speed and 0 for standstill. No reversing. There is also a analog speed display (servo motor), where the motor rotates to point to the current speed (same as a standard speedometer in cars). [Mandatory, 25 points]
3. The skateboard has a built-in fall detection sensor (accelerometer); if you're falling down (perhaps due an accident), the skateboard instantly stops. [Elective, 10 points]
4. The skateboard has another safety feature: obstacle detection. The skateboard can detect (ultrasound sensor) when it comes too close to an obstacle and can slows down automatically. The braking strength is proportional to the distance from the obstacle. [Elective, 10 points]
5. The remote also has a “drone” option, where you can control the skateboard from a distance. This is done through the joystick. One axis of the joystick controls the speed (all other functions, i.e., speed increase-decrease, speed display, etc. remains the same). The other axis is used to steer the skateboard (stepper motor, connected by the h-bridge, and not the stepper motor driver).

Pressing the joystick button activates the “drone mode”; when the drone mode is activated the accelerometer cannot be used for control anymore. Pressing the joystick again deactivates the “drone mode” and control is passed back to the accelerometer. [Mandatory, 25 points]

6. The skateboard has a lighting setup (represented by a single LED, connected through the shift register). The lighting is controlled jointly by a photoresistor (dark environment uses a bright light, well-lit environment turns off the light) and a potentiometer. If the potentiometer is kept at min position, then the brightness of the LED is controlled by the photoresistor. [Elective, 10 points]

Oral questions [Mandatory, 4 questions x 5 points each, 20 points]

Commented code [Mandatory, 10 points]

You will see that the total point obtainable are 120. Your final grade (out of 100) will be *max (100, points obtained)*. The mandatory functionalities you must demonstrate, you may choose to demonstrate some of the elective functionalities. It is highly recommended that you attempt all, thereby create a safety net in case something goes wrong during the demo or you don't do well in the oral part. During the demo, you will be able to use the breadboard power supply module to directly provide power to the breadboard. Your project needs to be completely ready by the beginning of the demo day; you will have 4 minutes each to demonstrate your project and answer oral questions. You will need to upload a detailed commented code. This is an individual project; interacting with or helping each other is allowed, but identical codes will not be accepted.

Recommendations: Develop the functionalities/components one by one. Once each functionality is independently working, bring them together. Use time outside of class hours to develop the code and use the class hours for hardware work. Aim to have the entire project completed by the beginning of class on 24 June, use the class time for final debugging, tidying up your breadboard, etc. Utilize office hours. Most importantly, enjoy getting to create something with your own hands.

Timeline:

14 June: Problem statement uploaded

17 June, 24 June: 1.5 hours class for preparing project

18 June, 20 June: office hours

19 June: No class, public holiday

26 June: Demo day